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Math, Fractions, Decimals, Ratios, Percentages & Vedic Speed Calculation

Complete Reference Guide, Illustrated Textbook & Cheat-Sheet Workbook with Step-by-Step Animated Visual Flows & Verified Academic Sources



Visual Architecture of Numbers: Geometric ratios, circular fraction models, golden spiral proportions, and rapid mental calculation pathways synthesized for intuitive mastery.

This comprehensive textbook and workbook unites foundational arithmetic—**Fractions, Decimals, Ratios, and Percentages**—with high-speed mental algorithms from **Vedic Speed Calculation**. Every concept is paired with frame-by-frame animated sequence diagrams, visual bar and grid models, rapid conversion tables, and graded workbook exercises grounded in authoritative curriculum texts.

Module	Core Focus & Speed Techniques	Primary Grounded Reference
Ch. 1–2: Fractions & Decimals	Part-whole visual models, LCM & butterfly addition, base-10 place value, repeating decimal algebra	OpenStax Prealgebra 2e (Ch. 4 & 5) & NCERT Mathematics
Ch. 3–4: Ratios & Percentages	Bar scaling, extremes-means cross law, Alligation cross-flow, 1/2 to 1/20 equivalence grid, successive % change	OpenStax (Ch. 5.6 & 6.1) & NCERT Fractions in Disguise
Ch. 5: Vedic Speed Math	All 16 Sutras & 13 Sub-Sutras, *Urdhva Tiryagbhyam* crosswise flow, *Nikhilam* base multiplication, *Ekadhikena* squaring	Bharati Krishna Tirthaji (1965) & VHU 16 Sutras Compendium
Ch. 6–7: Cheat-Sheet & Workbook	Consolidated master formula table, 15 graded speed-drill problems, complete worked solutions & bibliography	Khan Academy Rate, Ratio & Percent

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Chapter 1: Fractions Build the Foundation of Proportional Reasoning

As established in OpenStax Prealgebra 2e (Section 4.1: Visualize Fractions), a **fraction** is written in the form $\mathbf{a / b}$, where $\mathbf{a}$ (the **numerator**) counts how many equal parts are selected, and $\mathbf{b}$ (the **denominator**, $\mathbf{b \neq 0}$) defines the total number of equal parts into which the whole unit is divided. Because division by zero is undefined, the denominator can never be zero.

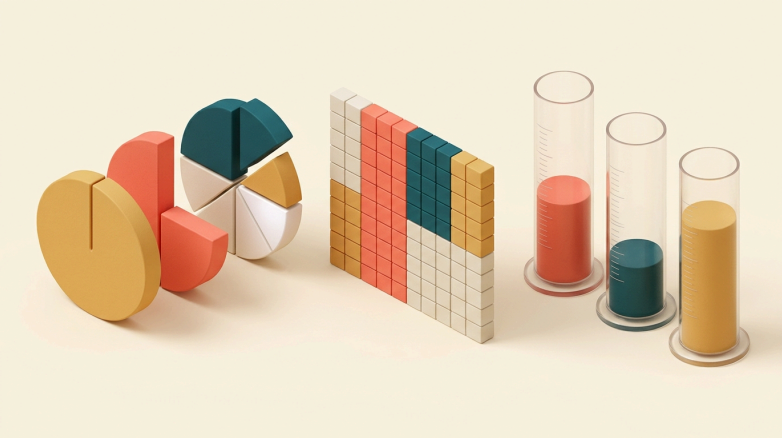


Figure 1.1 — 3D Equivalence Model: Segmented circular fraction wedges (parts of a circle), a 10×10 hundredths decimal/percent block, and graduated proportional cylinders demonstrating how fractions, decimals, and ratios represent identical physical quantities (Khan Academy Visual Model).

1.1 Taxonomy of Fractions & Mixed-Number Conversions

Following the classification in OpenStax Prealgebra 2e Section 4.1, every fraction belongs to one of three structural types, and mastering rapid two-way conversion between improper fractions and mixed numbers eliminates calculation bottlenecks:

- **Proper Fractions ($\mathbf{a < b}$):** The numerator is strictly smaller than the denominator, so the value lies between $\mathbf{0}$ and $\mathbf{1}$ on the number line (e.g., $\mathbf{3/4}$, $\mathbf{5/8}$, $\mathbf{11/18}$).
- **Improper Fractions ($\mathbf{a \geq b}$):** The numerator is greater than or equal to the denominator, representing a quantity greater than or equal to $\mathbf{1}$ whole (e.g., $\mathbf{11/6}$, $\mathbf{33/8}$, $\mathbf{7/3}$). By the **Property of One**, any non-zero number divided by itself equals one ($\mathbf{a/a = 1}$).
- **Mixed Numbers ($\mathbf{W + r/b}$):** Combines a whole number $\mathbf{W}$ and a proper fraction $\mathbf{r/b}$ (e.g., $\mathbf{1\ 5/6 = 11/6}$).
  Animated Cycle Rule (Mixed → Improper): Multiply down ($\mathbf{W \times b}$), add up ($\mathbf{+ r}$), and keep the denominator $\mathbf{b}$: for $\mathbf{4\ 2/3}$, compute $\mathbf{(4 \times 3 + 2) / 3 = 14/3}$.
  Animated Reverse Rule (Improper → Mixed): Divide $\mathbf{a \div b}$ to get Quotient $\mathbf{Q}$ and Remainder $\mathbf{R}$, then write $\mathbf{Q\ R/b}$: for $\mathbf{33/8}$, $\mathbf{33 \div 8 = 4}$ with remainder $\mathbf{1}$, yielding $\mathbf{4\ 1/8}$.

1.2 Equivalent Fractions & The Butterfly Cross-Multiplication Method

The **Equivalent Fractions Property** (OpenStax Section 4.1) states that multiplying or dividing both numerator and denominator by the same non-zero constant $\mathbf{c}$ preserves the exact value: $\mathbf{a/b = (a \cdot c) / (b}$

· c)**. For instance, **1/2 = 2/4 = 3/6 = 4/8 = 50/100**. When adding, subtracting, or comparing unlike fractions without computing a slow prime factorization, use the visual **Cross-Multiplication ("Butterfly") Motion Method**:

Operation	Algebraic Formula	Step-by-Step Animated Motion Guide	Worked Numerical Example
Rapid Comparison	Compare **a·d** vs. **b·c**	Shoot upward diagonal arrows from each denominator to the opposite numerator: **d ↗ a** and **b ↖ c**. The larger product marks the larger fraction.	Compare **5/8** and **7/11**: Left wing: **5 × 11 = 55** Right wing: **8 × 7 = 56** Since **55 < 56**, **5/8 < 7/11**.
Butterfly Addition	**a/b + c/d = (ad + bc) / bd**	Frame 1: Diagonal left **a·d**. Frame 2: Diagonal right **b·c**. Frame 3: Add wings over base product **b·d**.	**2/5 + 3/7**: **(2×7 + 5×3) / (5×7)** **= (14 + 15) / 35 = 29/35**.
Butterfly Subtraction	**a/b − c/d = (ad − bc) / bd**	Subtract the right diagonal wing from the left diagonal wing and place over **b·d**.	**5/6 − 2/5**: **(5×5 − 6×2) / (6×5)** **= (25 − 12) / 30 = 13/30**.
Multiplication & Cross-Cancel	**(a/b) × (c/d) = (a·c) / (b·d)**	First cancel any common factor diagonally between **a** and **d** or **c** and **b**, then multiply straight across.	**(9/14) × (7/15)**: Cancel **7** & **14** (→ **1, 2**); cancel **9** & **15** (→ **3, 5**) → **3/10**.
Reciprocal Division (Keep-Change-Flip)	**(a/b) ÷ (c/d) = (a/b) × (d/c)**	Keep the first fraction, change ÷ to ×, and flip the divisor into its reciprocal **d/c**.	**(3/4) ÷ (9/16)**: **= (3/4) × (16/9) = 4/3 = 1 1/3**.

Chapter 2: Decimals Extend Base-10 Place Value Beyond the Unit Point

As detailed in OpenStax Prealgebra 2e (Section 5.1: Decimals) and NCERT Mathematics (A Peek Beyond the Point), decimals are shorthand representations of fractions whose denominators are powers of ten (**10, 100, 1000, 10^n**). Moving one place to the right of the decimal point divides the place value by **10**: Tenths (**10^{-1} = 1/10 = 0.1**), Hundredths (**10^{-2} = 1/100 = 0.01**), Thousandths (**10^{-3} = 1/1000 = 0.001**), and Ten-Thousandths (**10^{-4} = 1/10000 = 0.0001**).

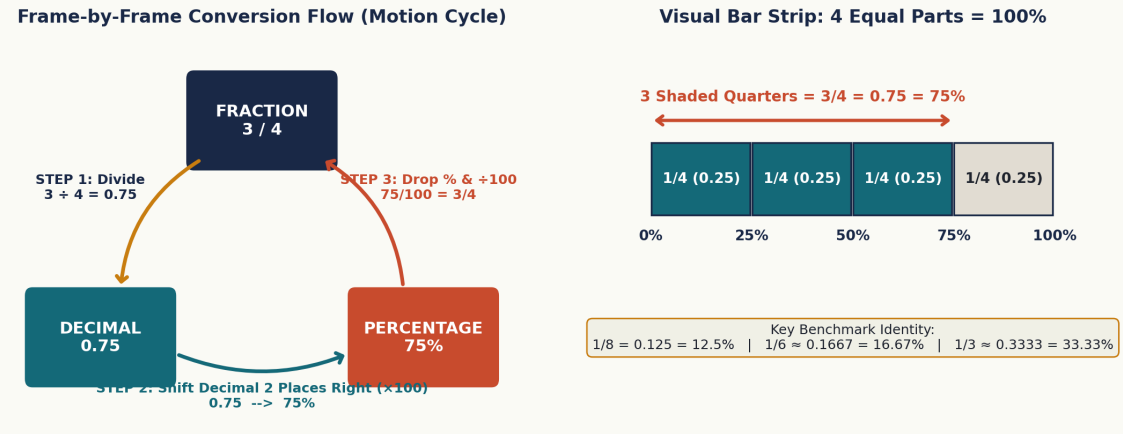


Figure 2.1 — Frame-by-Frame Conversion Motion Cycle: Left panel illustrates the three-step continuous transformation between Fraction ($3/4$), Decimal (0.75), and Percentage (75%). Right panel displays the aligned 4-part bar strip proving visual equivalence (OpenStax Section 6.1).

2.1 Terminating vs. Repeating Decimals & Instant Algebraic Conversion

Every rational number a/b (in simplest form) produces either a **terminating decimal** or a **repeating (recurring) decimal**:

- **Terminating Decimal Condition:** A reduced fraction a/b terminates if and only if the prime factorization of its denominator b contains only 2s and/or 5s ($b = 2^m \cdot 5^n$). To convert mentally without long division, pair every factor of 2 with a factor of 5 to build 10s in the denominator: e.g., $7/40 = 7 / (2^3 \cdot 5^1) = (7 \times 5^2) / 10^3 = 175 / 1000 = 0.175$.
- **Pure Repeating Decimal Shortcut (The 9s Rule):** Any decimal where a block of k digits repeats immediately after the decimal point equals that repeating block divided by k nines ($9, 99, 999$):
  • $0.777\ldots = 0.777\ldots = 7/9$ | $0.272727\ldots = 0.2727\ldots = 27/99 = 3/11$ | $0.142857\ldots = 142857/999999 = 1/7$.
- **Mixed Repeating Decimal Shortcut (The 9s and 0s Rule):** If m non-repeating digits precede k repeating digits after the decimal point, subtract the non-repeating part from the whole decimal digits and divide by k nines followed by m zeros:
  • Convert $0.1666\ldots = 0.1666\ldots$: $(16 - 1) / 90 = 15 / 90 = 1/6$.
  • Convert $0.23535\ldots = 0.23535\ldots$: $(235 - 2) / 990 = 233 / 990$.

2.2 High-Speed Decimal Arithmetic Rules

- **Addition & Subtraction (Vertical Point Alignment):** Pad trailing zeros so both numbers have identical decimal length, align the decimal points vertically, and add or subtract column by column (OpenStax Section 5.1).
- **Multiplication (Count & Shift Left):** Multiply the numbers as whole integers ignoring the decimal point, then count the total number of decimal places in both factors ($d_1 + d_2$) and shift the decimal point that many places from the right (e.g., $1.25 \times 0.04 \rightarrow 125 \times 4 = 500$; shift $2 + 2 = 4$ places left → $0.0500 = 0.05$).
- **Division (Equal Right-Shift Normalization):** If the divisor has k decimal places, shift the decimal point k places to the right in *both* divisor and dividend before dividing (e.g., $4.32 \div 0.12 = 432 \div 12 = 36$).

Chapter 3: Ratios, Proportions & Alligation Govern Relative Scaling

According to OpenStax Prealgebra 2e (Section 5.6: Ratios and Rate) and Khan Academy Rate, Ratio, Proportion and Percentage, a **ratio** compares two quantities of the same kind by division ($a : b$ or a/b, where $b \neq 0$), whereas a **rate** compares quantities measured in different units (such as $180\ \text{km} / 3\ \text{hours} = 60\ \text{km/h}$, the **unit rate** where the denominator is 1).

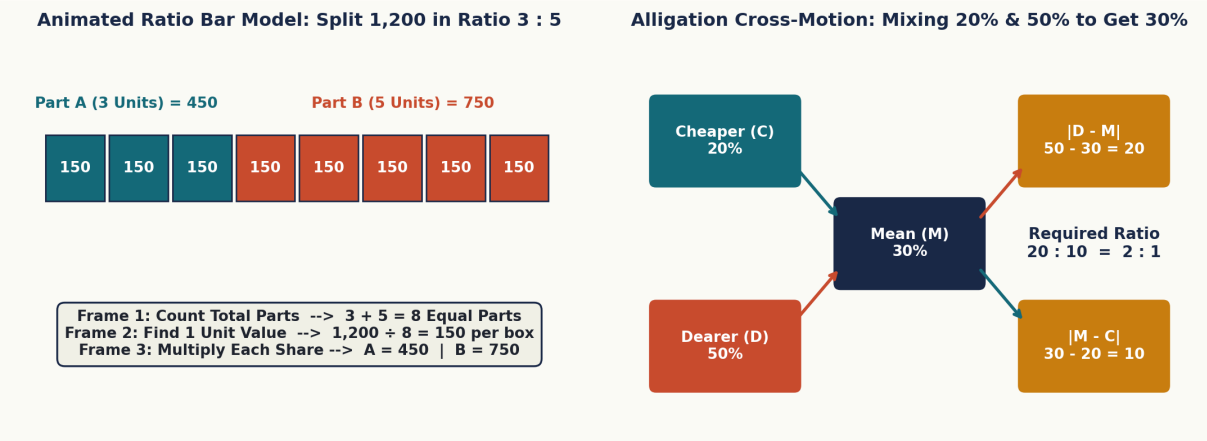


Figure 3.1 — Animated Ratio Bar Model & Alligation Cross-Flow: Left panel shows the 3-frame unit-box method for partitioning a total quantity ($1{,}200$ split in ratio $3{:}5$). Right panel illustrates the diagonal cross-subtraction arrows of the Alligation Rule for blending two concentrations.

3.1 Core Laws of Ratios, Proportions & Chain Merging

Law / Principle	Mathematical Identity	Speed Application & Worked Example
Part-to-Whole Partitioning	$\textbf{Share A} = \textbf{Total} \times a / (a + b)$ $\textbf{Share B} = \textbf{Total} \times b / (a + b)$	To divide $1{,}200$ in the ratio $3 : 5$: total parts = 8. One unit = $1200 \div 8 = 150$. Share A = 450, Share B = 750.
Cross-Product of Extremes & Means	If $a : b :: c : d$, then $a \cdot d = b \cdot c$	Solve $6 / x = 9 / 15$: cross-multiply $9x = 6 \times 15 = 90 \Rightarrow x = 10$.
Mean, Third & Fourth Proportional	Mean of a, b: $\sqrt{(ab)}$ Third of a, b: b^2/a	Mean proportional between 9 and 16 is $\sqrt{(9 \times 16)} = 12$. Third proportional to $4, 6$ is $36/4 = 9$.
Componendo & Dividendo	If $a/b = c/d$, then $(a + b)/(a - b) = (c + d)/(c - d)$	Instantly solves symmetric equations: if $(x + 3)/(x - 3) = 5/2$, then $x/3 = (5 + 2)/(5 - 2) = 7/3 \Rightarrow x = 7$.
Inverted N-Bridge Ratio Merging	If $A{:}B = a{:}b$ and $B{:}C = c{:}d$, $A : B : C = ac : bc : bd$	If $A{:}B = 2{:}3$ and $B{:}C = 4{:}5$, trace the N-arrows ($2{\times}4 : 3{\times}4 : 3{\times}5$) to get $A{:}B{:}C = 8 : 12 : 15$ in one step.

Chapter 4: Percentages Standardize Ratios to a Base of 100

As defined in **OpenStax Prealgebra 2e (Section 6.1: Understand Percent)** and **NCERT Mathematics (Fractions in Disguise)**, the word *percent* comes from the Latin *per centum* ("per one hundred"). A percentage is a ratio whose denominator is **100**: **p% = p / 100 = 0.01p**. Memorizing the reciprocal equivalence grid below transforms slow multi-digit percentage calculations into instant single-step fraction reductions.

4.1 Master Fraction–Decimal–Percentage Equivalence Grid

Unit Fraction	Exact Decimal	Exact Percentage	Derived Multiples (High-Frequency Speed Benchmarks)
1 / 2	0.50	50%	**3/2 = 1.5 = 150%** | **5/2 = 2.5 = 250%**
1 / 3	0.3333...	33.33% (33 1/3%)	**2/3 = 0.6667 = 66.67% (66 2/3%)** | **4/3 = 133.33%**
1 / 4	0.25	25%	**3/4 = 0.75 = 75%** | **5/4 = 1.25 = 125%**
1 / 5	0.20	20%	**2/5 = 40%** | **3/5 = 60%** | **4/5 = 80%**
1 / 6	0.1666...	16.67% (16 2/3%)	**5/6 = 0.8333 = 83.33% (83 1/3%)**
1 / 7	0.142857...	14.28% (14 2/7%)	**2/7 = 28.57%** | **3/7 = 42.86%** | **4/7 = 57.14%**
1 / 8	0.125	12.5% (12 1/2%)	**3/8 = 37.5%** | **5/8 = 62.5%** | **7/8 = 87.5%**
1 / 9	0.1111...	11.11% (11 1/9%)	**2/9 = 22.22%** | **4/9 = 44.44%** | **7/9 = 77.78%**
1 / 11	0.0909...	9.09% (9 1/11%)	**2/11 = 18.18%** | **3/11 = 27.27%** | **5/11 = 45.45%**
1 / 12 & 1 / 16	0.0833 / 0.0625	8.33% / 6.25%	**5/12 = 41.67%** | **1/15 = 6.67%** | **1/20 = 5%** | **1/25 = 4%**

4.2 High-Speed Percentage Techniques & Commercial Identities

- **The Commutative Swap Law (x% of y = y% of x):** Since **(x/100) × y = (y/100) × x**, swap awkward percentages immediately. To find **48% of 25**, flip to **25% of 48 = (1/4) × 48 = 12** in one second!
- **Split-and-Decompose Method:** Break any percentage into anchor blocks of **10%** (shift decimal 1 place left), **5%** (half of **10%**), and **1%** (shift decimal 2 places left). For **18% of 450**: compute **20% (90) – 2% (9) = 81**.
- **Successive Percentage Change Formula:** If a value changes successively by **A%** and then **B%**, the net single percentage change is **Net % = A + B + (A · B)/100** (**NCERT Comparing Quantities**). For two successive increases of **+20%** and **+10%**, Net = **20 + 10 + (200/100) = +32%**.

- **Constant-Product Offset Rule (Price × Consumption = Expenditure):** If a price rises by $R\%$ (or $1/n$), consumption must drop by $1/(n + 1)$ to keep expenditure unchanged. Example: If price increases by $25\% = 1/4$, reduce consumption by $1/(4 + 1) = 1/5 = 20\%$.

Chapter 5: Vedic Speed Calculation Transforms Arithmetic into One-Line Mental Flow

First compiled and published in 1965 by **Swami Bharati Krishna Tirthaji** in *Vedic Mathematics: Sixteen Simple Mathematical Formulae from the Vedas* (Motilal Banarsidass), this system organizes mental arithmetic around **16 concise Sanskrit aphorisms (Sutras)** and **13 corollaries (Sub-Sutras)** (VHU 16 Sutras & 13 Sub-Sutras Guide). While modern historians of mathematics—such as Prof. S. G. Dani (IIT Bombay) and K. S. Shukla—note that these rules are 20th-century algebraic and arithmetic shortcuts rather than ancient Vedic texts, their symmetry makes them exceptional mental-calculation accelerators for competitive exams and everyday numeracy.

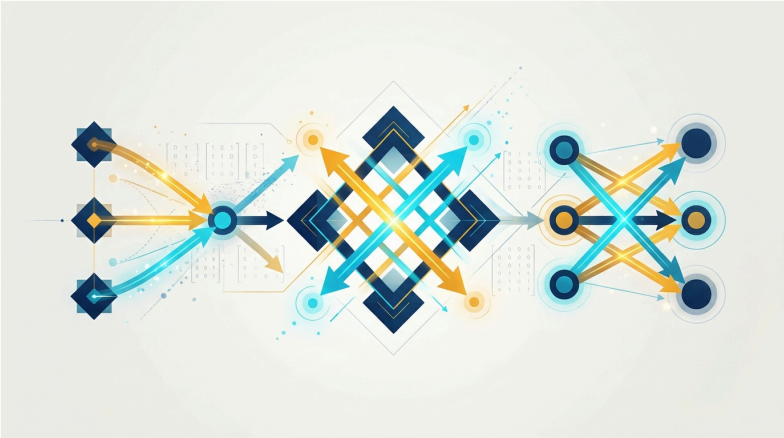


Figure 5.1 — Symmetric Criss-Cross Mental Pathways: Visual representation of how Vedic crosswise multiplication replaces multi-row partial products with a single symmetric left-to-right mental stream.

5.1 Complete Reference Table of the 16 Sutras of Vedic Mathematics

#	Sanskrit Sutra	English Meaning	Primary Speed Calculation Application
1	*Ekadhikena Purvena*	By one more than the previous one	Instant squaring of numbers ending in 5 ($85^2 = 7225$); recurring decimals for $1/19$, $1/29$
2	*Nikhilam Navatashcaramam Dashatah*	All from 9 and the last from 10	Instant subtraction from powers of 10; fast multiplication near bases 10, 100, 1000
3	*Urdhva Tiryagbhyam*	Vertically and crosswise	Universal 1-line multiplication for any 2×2, 3×3, or polynomial product

#	Sanskrit Sutra	English Meaning	Primary Speed Calculation Application
4	*Paravartya Yojayet*	Transpose and apply	Division when divisor is slightly above a base (e.g., **112, 103**) and linear equations
5	*Shunyam Samyasamuccaye*	When the sum is the same, that sum is zero	Instant solution of symmetric algebraic equations without expanding brackets
6	*Anurupye Shunyamanyat*	If one is in ratio, the other is zero	Solving simultaneous linear equations where one variable's coefficients are proportional
7	*Sankalana-Vyavakalanabhyam*	By addition and by subtraction	Solving symmetric systems ($\mathbf{ax + by = p}$, $\mathbf{bx + ay = q}$) by adding and subtracting once
8	*Puranapuranabhyam*	By the completion or non-completion	Completing the square/cube or grouping complements to **10** and **100** in mental sums
9	*Chalana-Kalanabhyam*	Differences and similarities	Factorizing quadratic and cubic expressions and differential root approximations
10	*Yaavadunam*	Whatever the extent of its deficiency	Instant squaring and cubing of numbers near a power of 10 (e.g., $\mathbf{96^2 = 9216}$)
11	*Vyashtisamanstih*	Part and whole	Finding weighted averages using deviations from an assumed mean
12	*Sheshanyankena Charamena*	The remainders by the last digit	Converting fractions into exact repeating decimals digit-by-digit
13	*Sopantyadvayamantyam*	The ultimate and twice the penultimate	Solving special reciprocal fraction equations in one inspection step
14	*Ekanyunena Purvena*	By one less than the previous one	Instant multiplication by strings of 9s ($\mathbf{47 \times 99 = 4653}$; $\mathbf{642 \times 999 = 641358}$)
15	*Gunitasamuchyah*	The product of the sum is equal to the sum of the product	Verifying factorization and polynomial multiplication via coefficient sums
16	*Gunakasamuchyah*	The factors of the sum is equal to the sum of the factors	Checking algebraic expansions and digital root (*Beejank*) verification

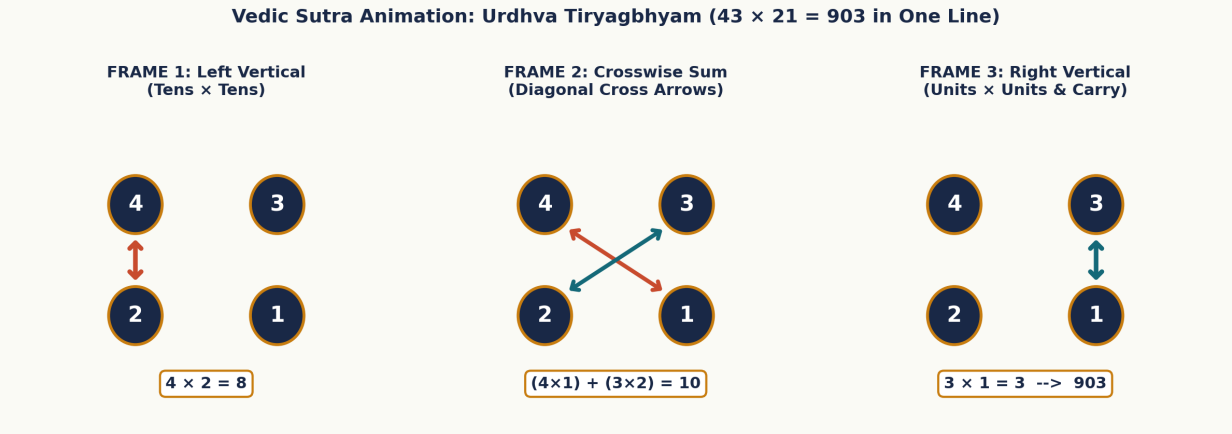


Figure 5.2 — 3-Frame Animated Storyboard for *Urdhva Tiryagbhyam* (43 × 21 = 903): Frame 1 multiplies the left vertical column ($4\times2=8$); Frame 2 sums the diagonal crosswise products ($4\times1 + 3\times2 = 10$, carry 1 to make 90); Frame 3 multiplies the right vertical column ($3\times1=3$) to yield 903.

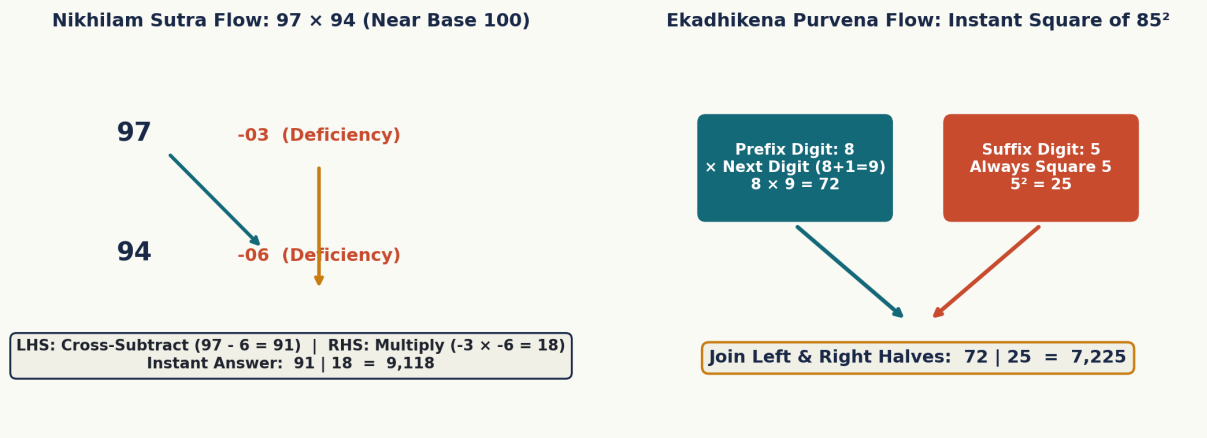


Figure 5.3 — Animated Motion Flow for *Nikhilam* (97 × 94) & *Ekadhikena Purvena* (85^2): Visual arrow paths show how two-second mental operations replace traditional multi-line multiplication.

5.2 Step-by-Step Animated Drills for the Top 6 Speed Sutras

1. **Sutra 1 — *Ekadhikena Purvena* (Squaring Numbers Ending in 5 & Complementary Pairs):**
   To square any number ending in 5 (form $\mathbf{N5}$), multiply the prefix $\mathbf{N}$ by $\mathbf{(N + 1)}$ and attach $\mathbf{25}$ on the right:
   • $\mathbf{65^2 \rightarrow [6 \times 7] \mid 25 = 4{,}225}$ | $\mathbf{115^2 \rightarrow [11 \times 12] \mid 25 = 13{,}225}$.
   Extension (Antyayordashake'pi): Whenever two numbers share the same tens prefix and their units digits add up to $\mathbf{10}$ (e.g., $\mathbf{43 \times 47}$), multiply $\mathbf{Prefix \times (Prefix + 1)}$ and attach the product of the unit digits: $\mathbf{43 \times 47 \rightarrow [4 \times 5] \mid [3 \times 7] = 2{,}021}$.

2. **Sutra 2A — *Nikhilam* Subtraction ("All from 9 and the Last from 10"):**
   To subtract any number from a power of $\mathbf{10}$ ($\mathbf{1{,}000; 10{,}000}$) with zero borrowing, subtract every digit from $\mathbf{9}$ and the final non-zero digit from $\mathbf{10}$:
   • $\mathbf{10{,}000 - 4{,}672 \rightarrow (9-4)(9-6)(9-7)(10-2) = 5{,}328}$.

3. **Sutra 2B — *Nikhilam* Base Multiplication (Below, Above & Mixed Base 100 / 1000):**
   • *Both Below Base 100:* Multiply $\mathbf{96 \times 92}$. Deficiencies from $\mathbf{100}$ are $\mathbf{-04}$ and $\mathbf{-08}$. Cross-subtract ($\mathbf{96 - 8 = 88}$) for the left half; multiply deficiencies ($\mathbf{(-4) \times (-8) = 32}$) for the two-digit right half → $\mathbf{8{,}832}$.
   • *Both Above Base 100:* Multiply $\mathbf{104 \times 107}$. Surpluses are $\mathbf{+04}$ and $\mathbf{+07}$. Cross-add ($\mathbf{104 + 7 = 111}$) and multiply surpluses ($\mathbf{4 \times 7 = 28}$) → $\mathbf{11{,}128}$.

• *One Above, One Below Base 100:* Multiply 104×97 ($+04$ and -03). Cross-combine ($104 - 3 = 101$, which represents $10{,}100$) and subtract $4 \times 3 = 12 \rightarrow 10{,}100 - 12 = 10{,}088$.

4. **Sutra 3 — *Urdhva Tiryagbhyam* (3×3 Star-Pattern Multiplication in One Line):**
   For $(a\ b\ c) \times (d\ e\ f)$, execute 5 symmetric motion frames:
   Frame 1 (Left Vertical): $a \cdot d$ | Frame 2 (Left Cross): $a \cdot e + b \cdot d$ | Frame 3 (3-Way Star): $a \cdot f + c \cdot d + b \cdot e$ | Frame 4 (Right Cross): $b \cdot f + c \cdot e$ | Frame 5 (Right Vertical): $c \cdot f$.
   • Example 123×321: Frame 1: $1 \times 3 = 3$; Frame 2: $1 \times 2 + 2 \times 3 = 8$; Frame 3: $1 \times 1 + 3 \times 3 + 2 \times 2 = 14$; Frame 4: $2 \times 1 + 3 \times 2 = 8$; Frame 5: $3 \times 1 = 3$. Combining with carries: $3 \mid 8 \mid {}_{1}4 \mid 8 \mid 3 = 39{,}483$.

5. **Sutra 10 — *Yaavadunam* (Duplex & Base Squaring in 2 Seconds):**
   To square a number near base 100 with deviation d, compute $(\text{Number} + d) \mid d^2$:
   • 94^2 ($d = -6$): $(94 - 6) \mid (-6)^2 = 88 \mid 36 = 8{,}836$.
   • 108^2 ($d = +8$): $(108 + 8) \mid 8^2 = 116 \mid 64 = 11{,}664$.

6. **Sutra 14 — *Ekanyunena Purvena* (Instant Multiplication by 99, 999, 9999):**
   To multiply any number N by an equal-length string of 9s, write $(N - 1)$ on the left and its *Nikhilam* complement on the right:
   • $73 \times 99 \rightarrow (73 - 1) \mid (99 - 72) = 72 \mid 27 = 7{,}227$.
   • $458 \times 999 \rightarrow (458 - 1) \mid (999 - 457) = 457 \mid 542 = 457{,}542$.

Chapter 6: Rapid Formula Cheat-Sheet, Speed Workbook & Complete Answer Key

Use the consolidated master formula table below for last-minute revision, then test your speed and accuracy on the **15 Graded Workbook Drills** before checking the step-by-step solutions.

6.1 Consolidated One-Page Master Formula Cheat-Sheet

Topic Domain	Core Identity / Speed Shortcut	Instant Verification / Example
Fraction Cross-Sum	$a/b \pm c/d = (ad \pm bc) / bd$	$3/4 + 2/5 = (15 + 8)/20 = 23/20 = 1.15$
Recurring Decimal → Fraction	$0.abab\ldots = ab / 99$ | $0.abbb\ldots = (ab - a) / 90$	$0.4545\ldots = 45/99 = 5/11$ | $0.1666\ldots = 15/90 = 1/6$
Componendo-Dividendo	$a/b = c/d \Rightarrow (a+b)/(a-b) = (c+d)/(c-d)$	$7/3 \Rightarrow (7+3)/(7-3) = 10/4 = 5/2$
Alligation Mixture Rule	$\text{Qty(Cheap)} / \text{Qty(Dear)} = (D - M) / (M - C)$	Mix 15% and 40% for $25\% \rightarrow (40-25):(25-15) = 3:2$
Successive Net % Change	$\text{Net \%} = A + B + (AB / 100)$	$+20\%$ then $-20\% \rightarrow 20 - 20 - 400/100 = -4\%$
Multiply by 5, 25, 125	$\times 5 = \times 10/2$ | $\times 25 = \times 100/4$ | $\times 125 = \times 1000/8$	$64 \times 125 = 64{,}000 \div 8 = 8{,}000$

Topic Domain	Core Identity / Speed Shortcut	Instant Verification / Example
Multiply by 11 (Add Neighbors)	**(a b c) × 11 = a | (a+b) | (b+c) | c**	**352 × 11 = 3 | (3+5) | (5+2) | 2 = 3,872**
Digital Root Check (*Beejank*)	Sum digits modulo 9: **DR(A × B) = DR(DR(A) × DR(B))**	**43 × 21 = 903: DR(43)=7, DR(21)=3, DR(7×3)=3 = DR(903)**

6.2 Speed-Drill Practice Workbook (15 Graded Problems)

#	Category	Problem Statement	Recommended Speed Method
P1	Fractions	Compute **4/7 + 3/8** without finding prime factorizations.	Butterfly Cross-Multiplication (**ad + bc** over **bd**)
P2	Fractions	Simplify **(14/25) ÷ (7/15)** to a mixed number.	Reciprocal Keep-Change-Flip & Diagonal Cancellation
P3	Fractions	Which fraction is larger: **9/13** or **11/16**?	Upward Diagonal Cross-Product Comparison
P4	Decimals	Convert the pure repeating decimal **0.363636... (repeating 36) 0.363636...** to its simplest fraction.	Pure Repeating 99s Rule
P5	Decimals	Convert the mixed repeating decimal **0.4666... (repeating 6) 0.4666...** to its simplest fraction.	Mixed Repeating 90s Subtraction Rule
P6	Decimals	Evaluate **0.048 ÷ 0.0016** mentally.	4-Place Right-Shift Decimal Normalization
P7	Ratios	If **A : B = 3 : 4** and **B : C = 6 : 7**, find the combined ratio **A : B : C**.	Inverted N-Bridge or Common Middle Term Scaling
P8	Ratios	In what ratio must a **12%** saline solution be mixed with a **32%** solution to obtain **20%**?	Alligation Cross-Subtraction Flow
P9	Percentages	Compute **37.5% of 640 + 16.67% of 420** mentally.	Master Equivalence Grid (**3/8** and **1/6**)
P10	Percentages	A jacket's price is raised by **25%** and later discounted by **20%**. Find the net percentage change.	Successive Percentage Formula **A + B + AB/100**
P11	Vedic Math	Find the exact square of 95^2 and the product **72 × 78** in 3 seconds.	Sutra 1: *Ekadhikena Purvena* & *Antyayordashake'pi*
P12	Vedic Math	Multiply **93 × 96** and **106 × 109** mentally.	Sutra 2: *Nikhilam* Base-100 Deficiency & Surplus

#	Category	Problem Statement	Recommended Speed Method
P13	Vedic Math	Compute 34×52 in a single line from left to right.	Sutra 3: *Urdhva Tiryagbhyam* (Vertically & Crosswise)
P14	Vedic Math	Compute $10{,}000 - 6{,}483$ and 584×999 instantaneously.	Sutra 2 (All from 9, last from 10) & Sutra 14 (*Ekanyunena*)
P15	Vedic Math	Compute 988^2 (near base $1{,}000$) in one line.	Sutra 10: *Yaavadunam* Base Deficiency Squaring

6.3 Complete Step-by-Step Worked Answer Key

- **P1 Solution:** $(4\times8 + 7\times3) / (7\times8) = (32 + 21) / 56 = 53/56$.
- **P2 Solution:** $(14/25) \times (15/7) = (2/5) \times (3/1) = 6/5 = 1\ 1/5$.
- **P3 Solution:** Left wing $9 \times 16 = 144$; Right wing $13 \times 11 = 143$. Since $144 > 143$, **9/13 > 11/16**.
- **P4 Solution:** $0.363636\ldots$ **(repeating 36)** $36 / 99$. Divide numerator and denominator by $9 \rightarrow$ **4/11**.
- **P5 Solution:** $0.4666\ldots$ **(repeating 6)** $(46 - 4) / 90 = 42 / 90$. Divide by $6 \rightarrow$ **7/15**.
- **P6 Solution:** Shift decimal 4 places right in both numbers: $480 \div 16 = 30$.
- **P7 Solution:** Using N-Bridge: $A:B:C = (3\times6) : (4\times6) : (4\times7) = 18 : 24 : 28$. Divide by $2 \rightarrow$ **9 : 12 : 14**.
- **P8 Solution:** By Alligation: $Qty(12\%) : Qty(32\%) = (32 - 20) : (20 - 12) = 12 : 8 = 3 : 2$.
- **P9 Solution:** $37.5\% = 3/8$ and $16.67\% = 1/6$. Thus $(3/8)\times640 + (1/6)\times420 = 240 + 70 = 310$.
- **P10 Solution:** $\text{Net \%} = +25 - 20 + (25 \times -20)/100 = 5 - 5 = 0\%$ **(No net change)**.
- **P11 Solution:** $95^2 = (9\times10) \mid 25 = 9{,}025$. For 72×78: $(7\times8) \mid (2\times8) = 5{,}616$.
- **P12 Solution:** 93×96 $(-07, -04)$: $(93-4) \mid (7\times4) = 8{,}928$. For 106×109 $(+06, +09)$: $(106+9) \mid (6\times9) = 11{,}554$.
- **P13 Solution:** Left vertical $3\times5=15$; Crosswise $(3\times2 + 4\times5) = 26$; Right vertical $4\times2=8$. Combine $15 \mid {}_{2}6 \mid 8 = 1{,}768$.
- **P14 Solution:** $10{,}000 - 6{,}483 = (9-6)(9-4)(9-8)(10-3) = 3{,}517$. For 584×999: $(584-1) \mid (999-583) = 583{,}416$.
- **P15 Solution:** Base = $1{,}000$, deficiency $d = -012$. Left part = $988 - 12 = 976$; Right part (3 digits) = $(-12)^2 = 144 \rightarrow$ **976,144**.

Chapter 7: Verified Bibliography & Real Academic Sources

Every mathematical definition, visual model, equivalence identity, and historical note in this EPICHUB publication is grounded in the following authoritative open-access textbooks and scholarly sources:

1. **OpenStax — Rice University (Fractions & Mixed Numbers):** Marecek, L., Anthony-Smith, M., & Mathis, A. H. (2020). *Prealgebra 2e, Section 4.1: Visualize Fractions*. Houston, TX: OpenStax. Direct source: https://openstax.org/books/prealgebra-2e/pages/4-1-visualize-fractions.

2. **OpenStax — Rice University (Base-10 Decimals & Arithmetic):** Marecek, L., et al. (2020). *Prealgebra 2e, Section 5.1: Decimals*. Houston, TX: OpenStax. Direct source: **https://openstax.org/books/prealgebra-2e/pages/5-1-decimals**.
3. **OpenStax — Rice University (Ratios & Unit Rates):** Marecek, L., et al. (2020). *Prealgebra 2e, Section 5.6: Ratios and Rate*. Houston, TX: OpenStax. Direct source: **https://openstax.org/books/prealgebra-2e/pages/5-6-ratios-and-rate**.
4. **OpenStax — Rice University (Percentages & Three-Way Conversions):** Marecek, L., et al. (2020). *Prealgebra 2e, Section 6.1: Understand Percent*. Houston, TX: OpenStax. Direct source: **https://openstax.org/books/prealgebra-2e/pages/6-1-understand-percent**.
5. **NCERT — National Council of Educational Research and Training (Decimals & Percentages):** NCERT Class 7 & Class 8 Mathematics Textbooks: *A Peek Beyond the Point* (**https://ncert.nic.in/textbook/pdf/gegp103.pdf**), *Fractions in Disguise* (**https://ncert.nic.in/textbook/pdf/hegp201.pdf**), and *Comparing Quantities Exemplar* (**https://ncert.nic.in/pdf/publication/exemplarproblem/classVIII/mathematics/heep209.pdf**).
6. **Khan Academy (Visual Equivalence Models & Rate/Ratio Curricula):** *Fraction, Decimal, and Percent from Visual Model* (**Khan Academy Pre-Algebra**) and *Unit 4: Rate, Ratio, Proportion and Percentage* (**Khan Academy Grade 7 Math**).
7. **Vedic Speed Calculation — Primary Treatise & Scholarly Historiography:** Tirthaji, Swami Bharati Krishna (1965). *Vedic Mathematics: Sixteen Simple Mathematical Formulae from the Vedas* (ed. V. S. Agrawala). Varanasi: Motilal Banarsidass (ISBN 978-8120801646). Documented with scholarly analysis by S. G. Dani (IIT Bombay) and K. S. Shukla at **Wikipedia: Vedic Mathematics** and the complete sutra reference at **Vedic Hindu University: 16 Sutras and 13 Sub-Sutras**.

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